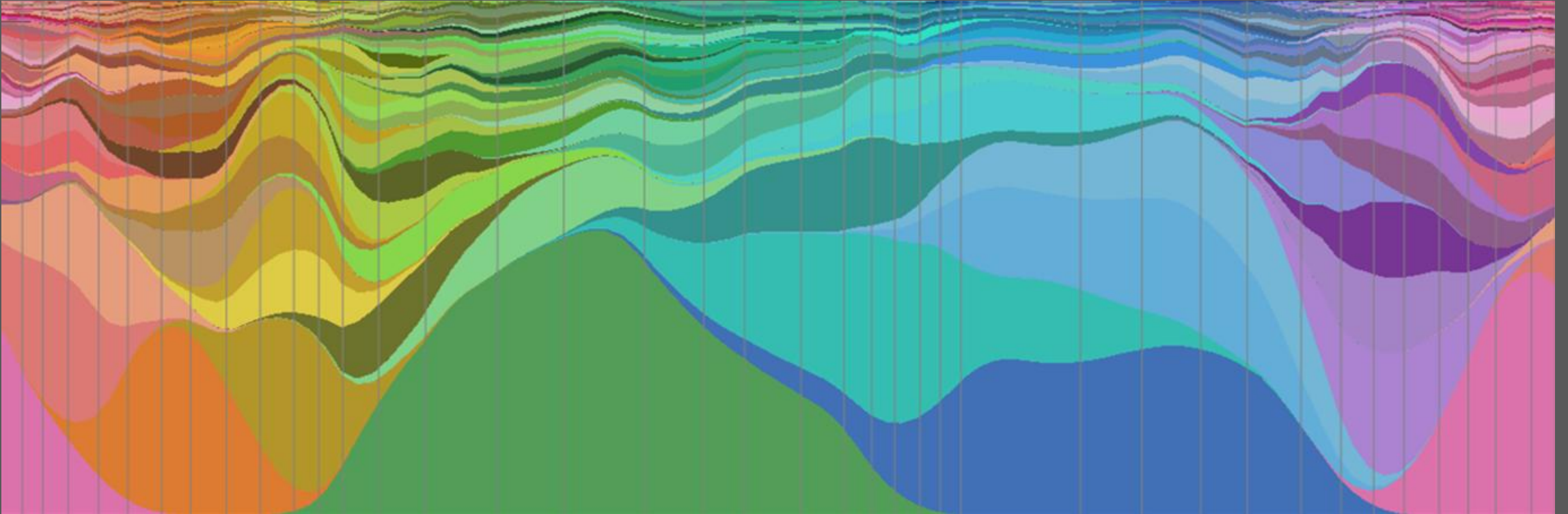


LDSCI5209 - Information Presentation and Visualisation

Week 12: Evaluation and Storytelling with Data



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Northeastern University London

Recap: Visualisation for machine learning

- Data exploration and preprocessing
- High-dimensional data
- Feature engineering
- Model performance and hyperparameter tuning

Plan for today

Evaluation and storytelling with data

- Evaluation and validation
- How to tell a story with data
- How to give a talk

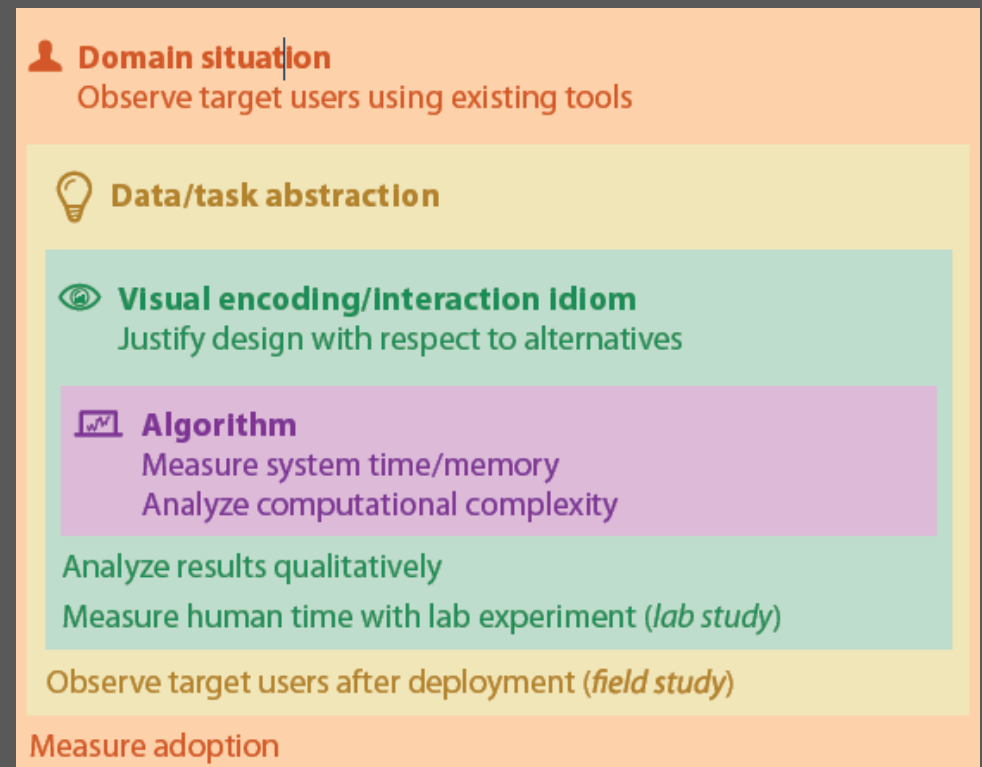
Validation and evaluation

Why, models, threats, process, criteria, techniques, usability

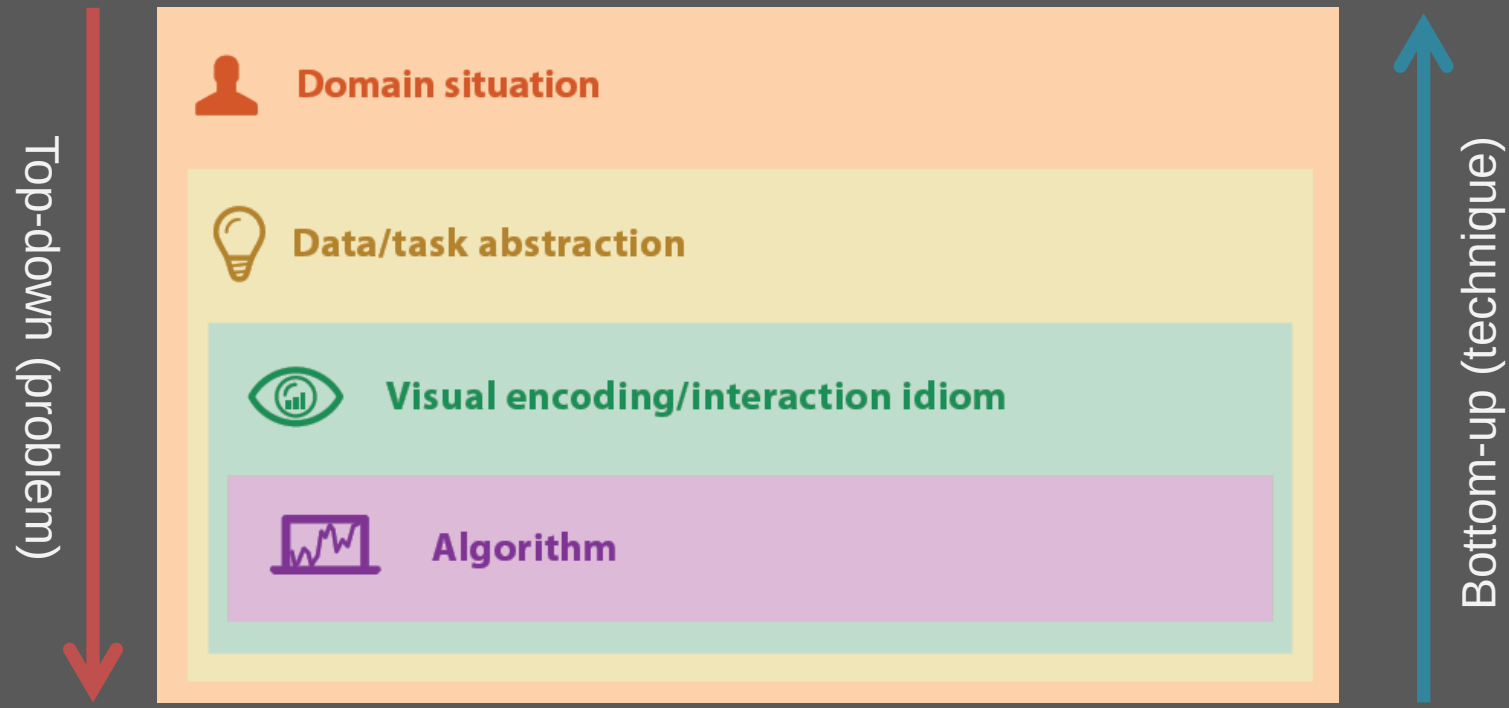
Why do we evaluate data visualisations?

- Understanding the audience
- Ensure clarity
- Optimise design
- Proof of effectiveness

Nested Model by Munzner (2009)

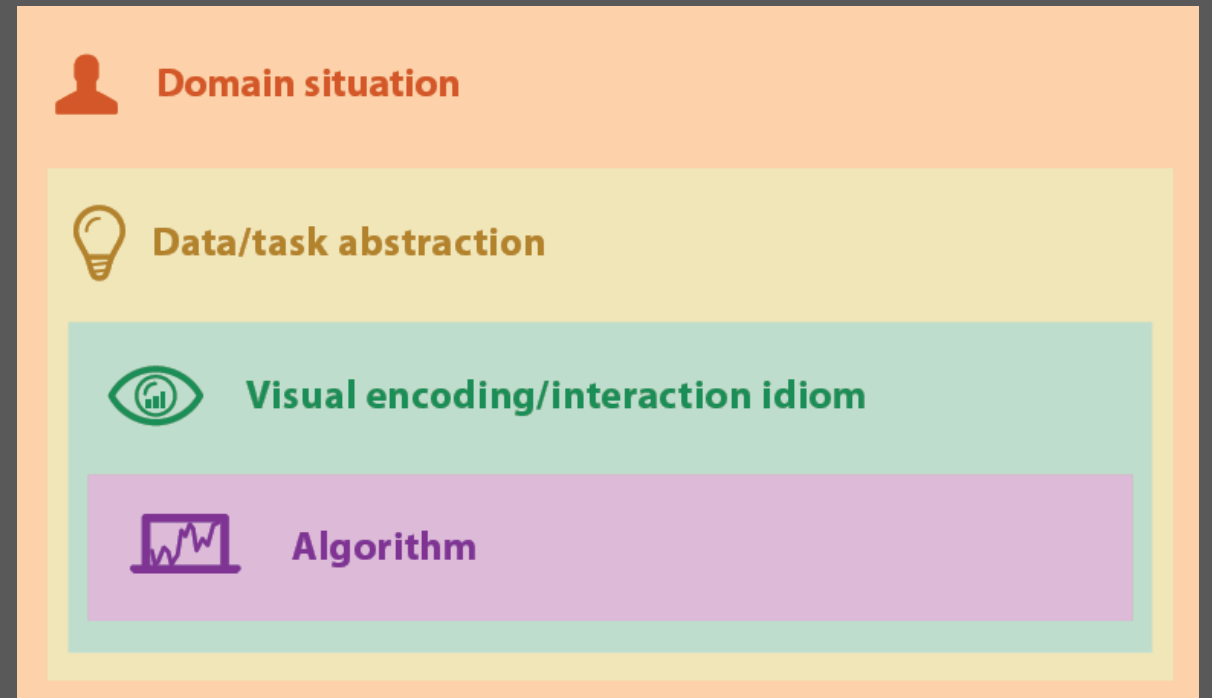


Nested model



Threats to validity

- Misunderstood user needs
- Show the wrong thing
- The presentation is ineffective
- Code is too slow



Threats to validity

❗ **Threat** Wrong problem

✅ **Validate** Observe and interview target users

❗ **Threat** Wrong task/data abstraction

❗ **Threat** Ineffective encoding/interaction idiom

✅ **Validate** Justify encoding/interaction design

❗ **Threat** Slow algorithm

✅ **Validate** Analyze computational complexity

Implement system

✅ **Validate** Measure system time/memory

✅ **Validate** Qualitative/quantitative result image analysis

Test on any users, informal usability study

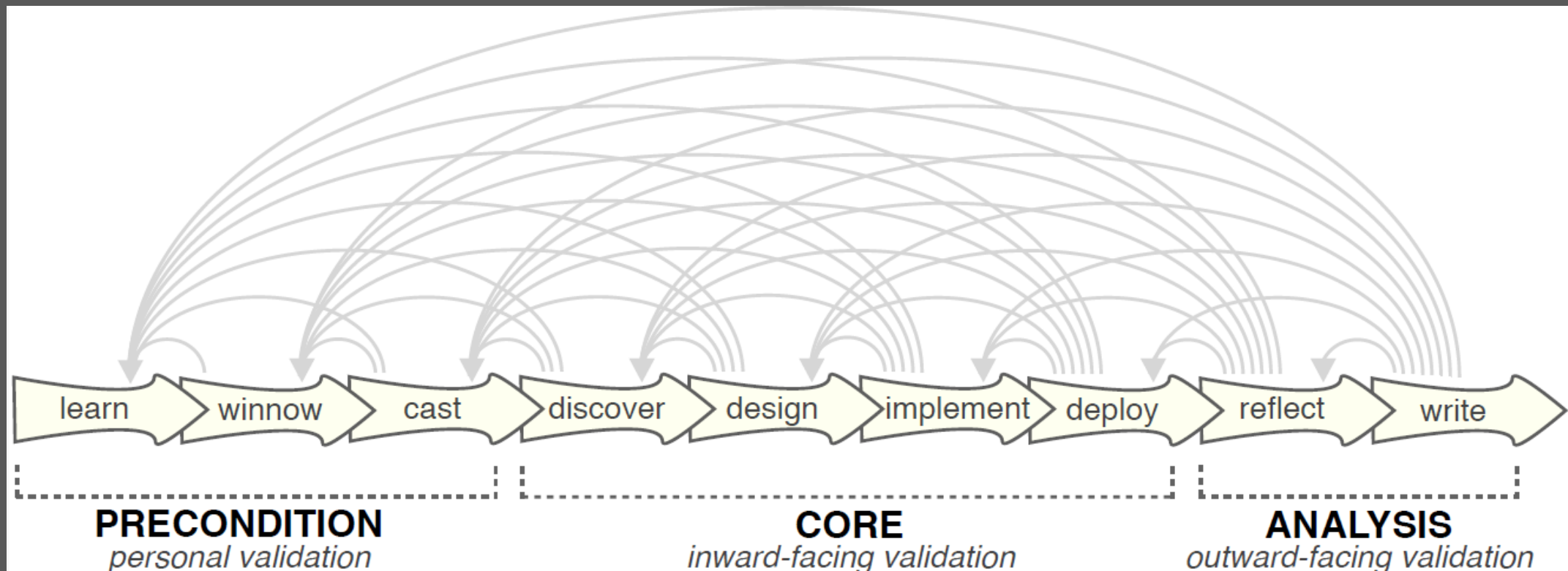
✅ **Validate** Lab study, measure human time/errors for task

✅ **Validate** Test on target users, collect anecdotal evidence of utility

✅ **Validate** Field study, document human usage of deployed system

✅ **Validate** Observe adoption rates

Validation process framework



Sedlmair et al., (2012): <http://www.cs.ubc.ca/labs/imager/tr/2012/dsm/>

Evaluation criteria

- **Functionality:** To what extent does the system provide the functionalities required by the users?
- **Effectiveness:** Does the visualization provide value? Does it provide new insight? How? Why?
- **Efficiency:** To what extent does the visualization help the users in achieve a better performance?
- **Usability:** How easily can the users interact with the system? Are the information provided in clear and understandable format?
- **Usefulness:** Are the visualization useful? How may we benefit from it?

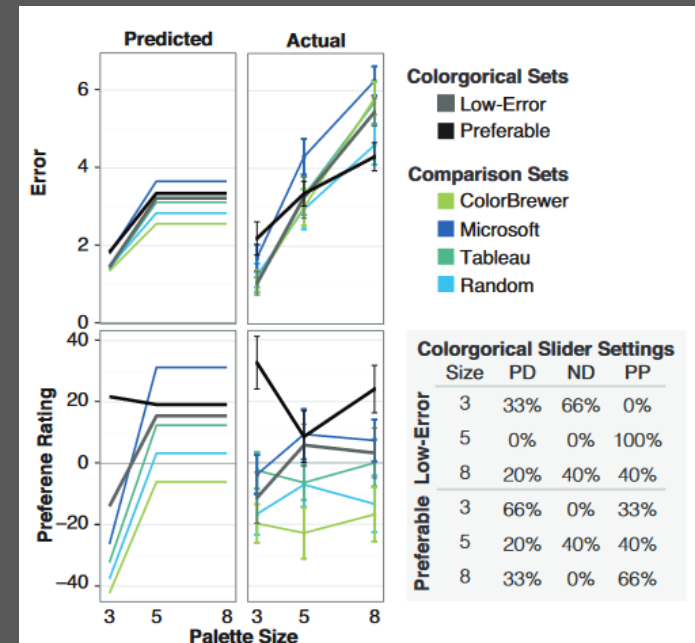
Evaluation Techniques

- **Quantitative** studies involve testing hypotheses through direct measurements, often using *controlled experiments* to manipulate factors and assess user performance.
- **Qualitative** studies analyse data from questionnaires, interviews, or observations to understand social phenomena. This includes *focus groups* for group discussions and *interviews/case studies* to gather user impressions and feedback.

Usability study of final project

Conduct usability testing of your nearly-complete final project visualisations considering:

- Relevance
- Clarity
- Design
- Efficiency



Gramazio et al., (2017)
<http://vrl.cs.brown.edu/color>

Storytelling

Narrative elements, scenarios, genres, examples

What is a story in visualisation?

- **Story:** an ordered sequence of steps with a clearly defined path through it.
- Steps can contain text, images, visualizations, or video, or combinations.
- Order is a key element of stories (usually linear in time).

Presenting a data story visually

Actions

Analyze

→ Consume

→ Discover



→ Present



→ Enjoy



→ Produce

→ Annotate



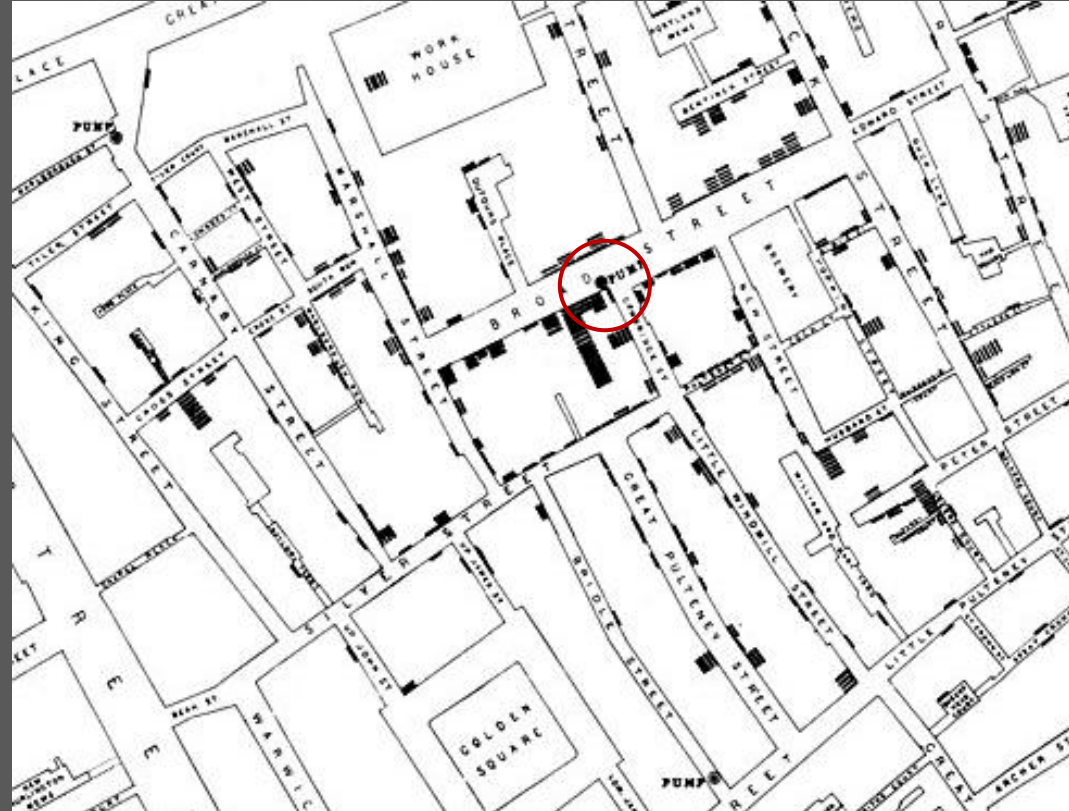
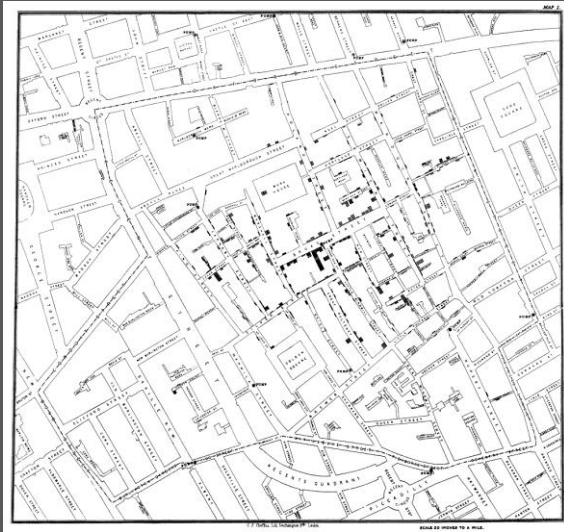
→ Record



→ Derive



Communicating how Cholera spreads



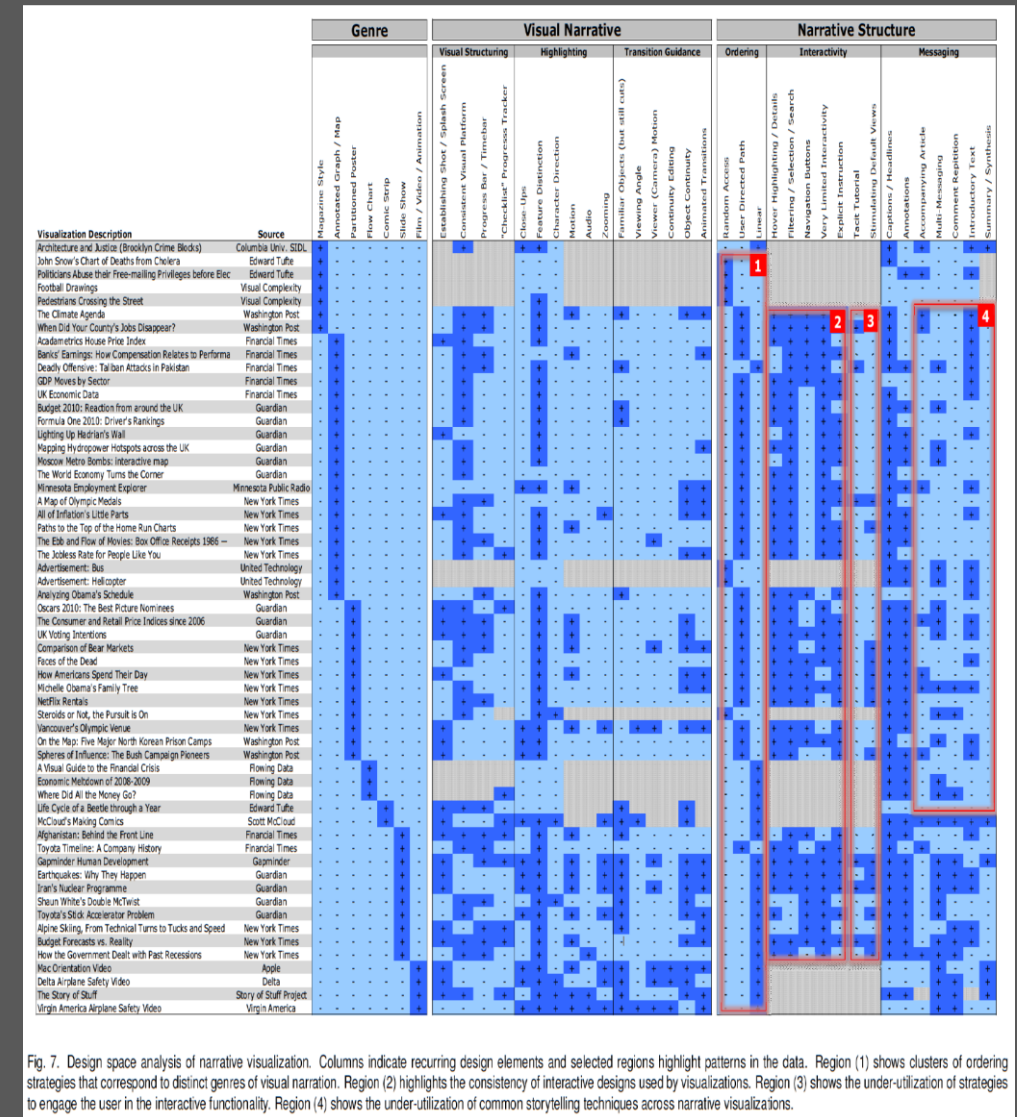
John Snow, 1854

Scenarios of storytelling

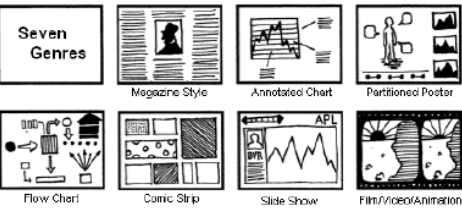
- **Self-running presentation to a large audience:** Self-explanatory; limited interaction; similar to a written story.
- **Live presentation by a speaker in front of audience:** Speaker can respond to audience, answer questions, pause story and interact. Additional layers/annotations, interactivity.
- **Individual/small-group presentation of results:** More interaction with audience; presentation leads to new questions; collect and consolidate information.

Narrative storytelling

- Visual Narrative techniques:
 - Visual Structure
 - Highlighting
 - Transition Guidance
- Narrative structures:
 - Ordering
 - Interactivity
 - Messaging



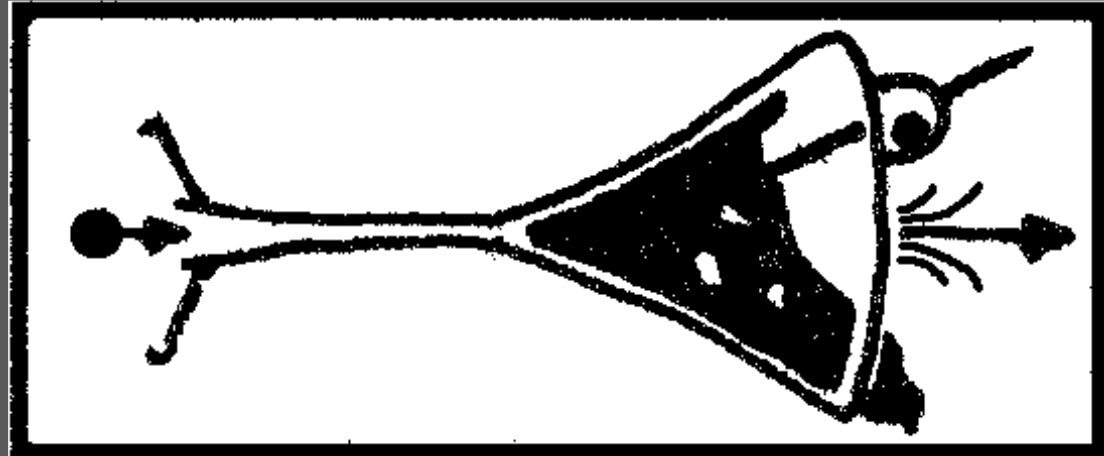
Genres of narrative data storytelling

Genre		Narrative Structure		
Visual Structuring	Visual Narrative	Ordering		Messaging
		Interactivity		
		Transition Guidance		
Magazine Style		Random Access		
Annotated Graph / Map		User Directed Path		
Partitioned Poster		Linear		
Flow Chart		Hover Highlighting / Details		
Comic Strip		Filtering / Selection / Search		
Slide Show		Navigation Buttons		
Film / Video / Animation		Very Limited Interactivity		
		Explicit Instruction		
		Tacit Tutorial		
		Stimulating Default Views		
		Captions / Headlines		
		Annotations		
		Accompanying Article		
		Multi-Messaging		
		Comment Repitition		
		Introductory Text		
		Summary / Synthesis		

Author- vs. Reader- driven structures

- Author-Driven
 - Linear ordering of scenes
 - Heavy messaging
 - No interactivity
- Reader-Driven
 - No prescribed ordering
 - No messaging
 - Free interactivity

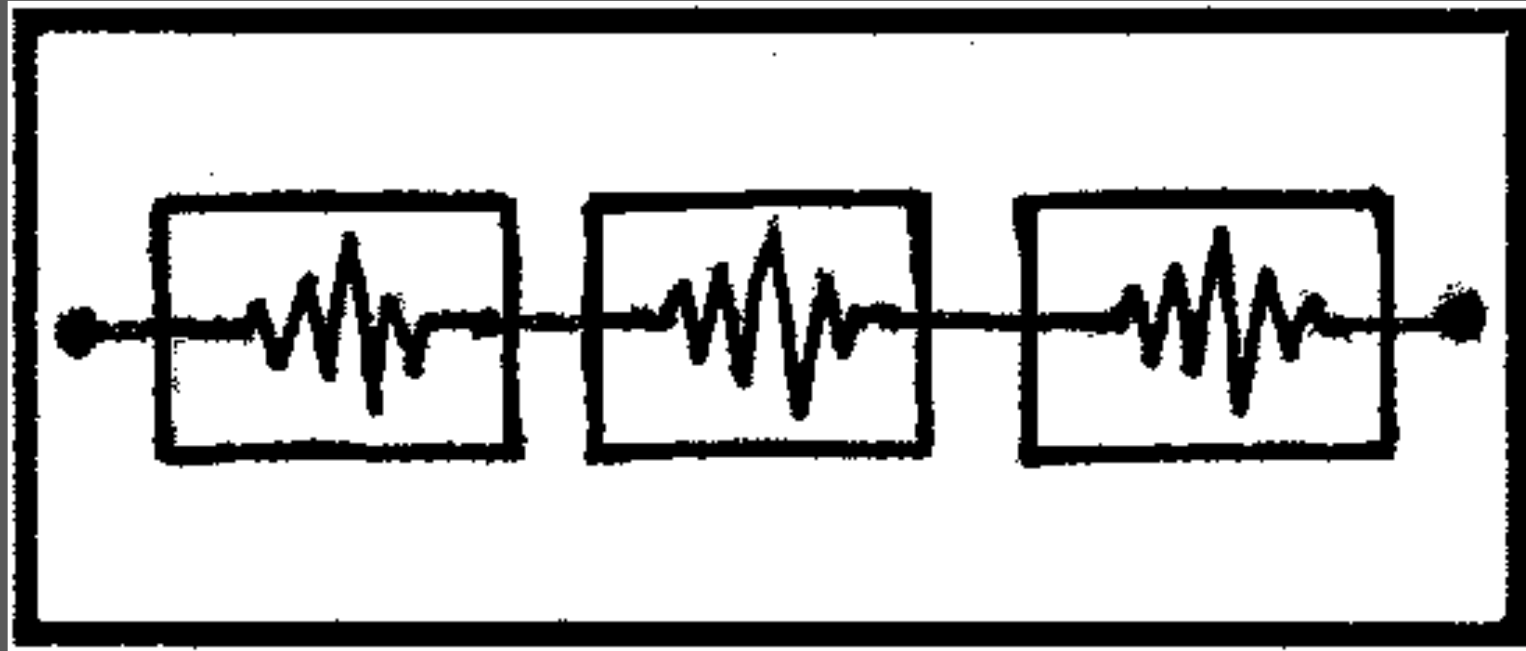
Starts as Author-driven



Leads to Reader-driven exploration

Author- vs. Reader- driven structures

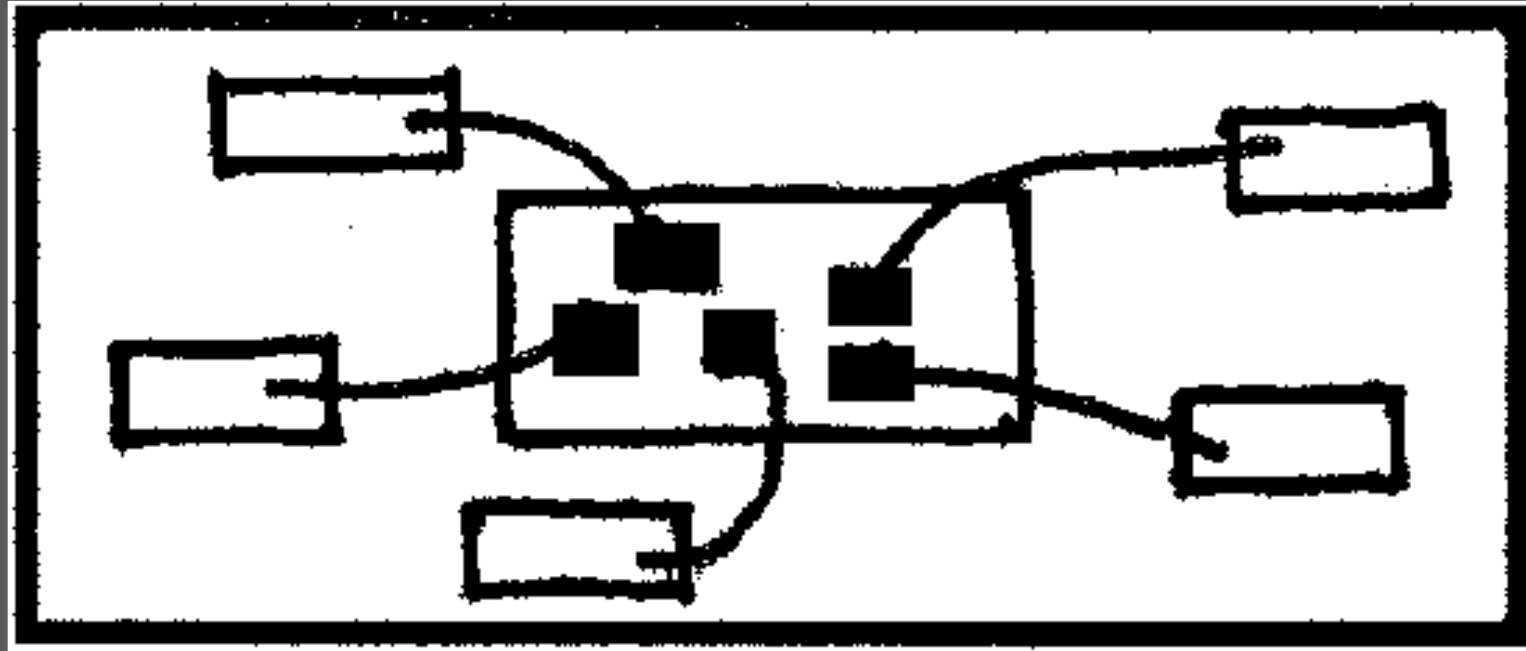
Interactive slideshow (e.g. Gapminder)



Overall Author-driven but allows Reader-driven exploration on each slide

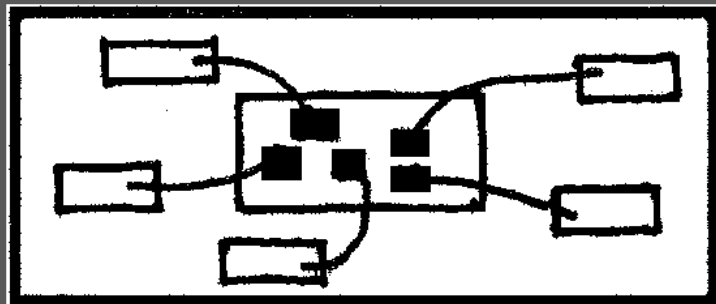
Author- vs. Reader- driven structures

Drill-down story



Overall Reader-driven but requires advanced Authoring to create narrative paths

Drill-down story



October 11, 2008

How This Bear Market Compares

[SIGN IN TO E-MAIL OR SAVE THIS](#) | [FEEDBACK](#)

The current bear market is already among the worst in history. Here is how it lines up – in losses and length – with those of the last 80 years.

The S. & P. 500, adjusted for inflation, shown monthly during bear markets.

Real S. & P. 500
Shaded area at right



Click on the lines to investigate each bear market

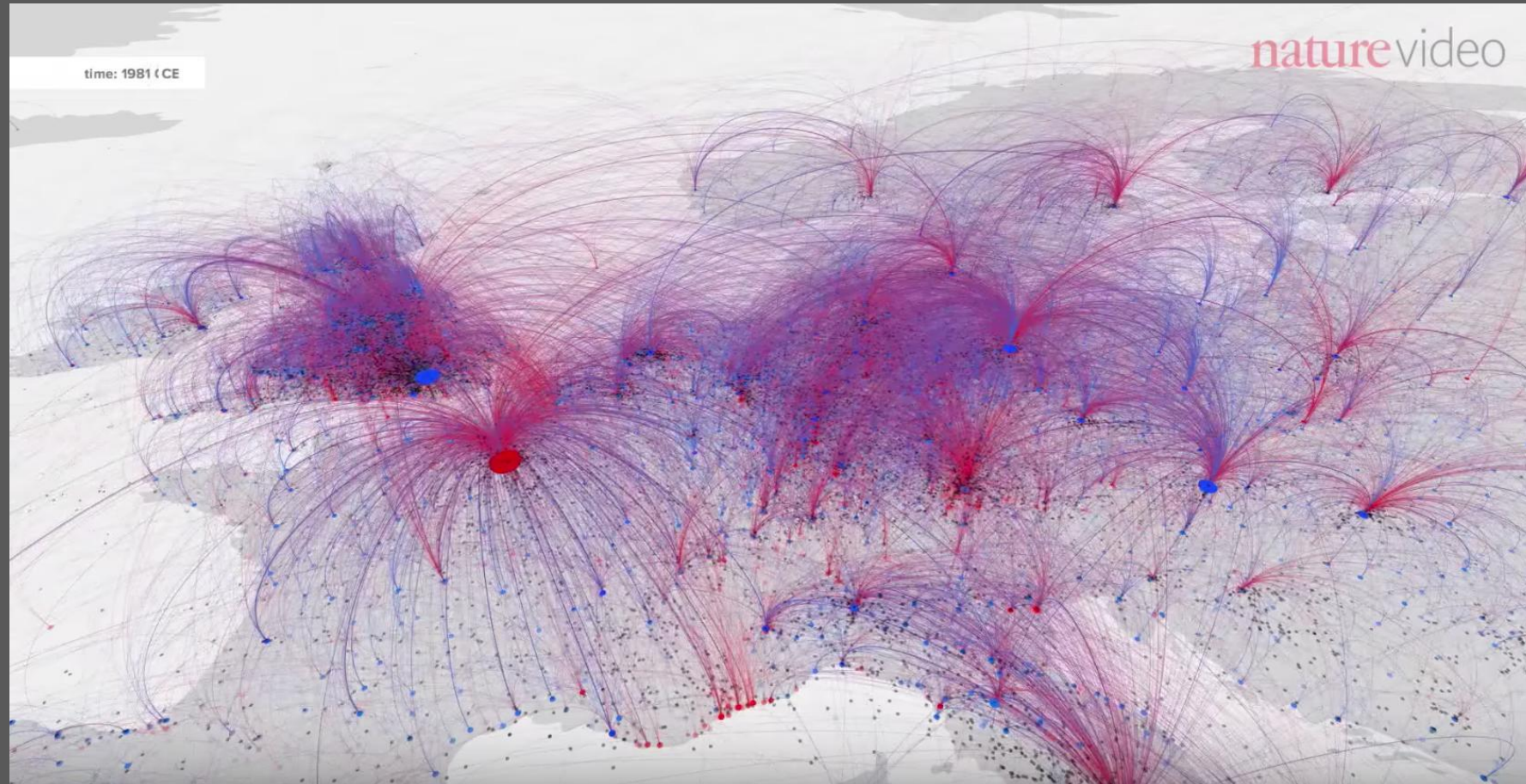


Stocks dropped 24 percent on Black Monday and Tuesday in 1929. And as bank failures began to mount and the economy sunk into a depression, they turned down again, sharply. At their low point, they were more than 80 percent below their 1929 high. After hitting bottom in 1932, it took 29 years - until 1958 - for the market to reach its pre-Depression, inflation-adjusted peak.

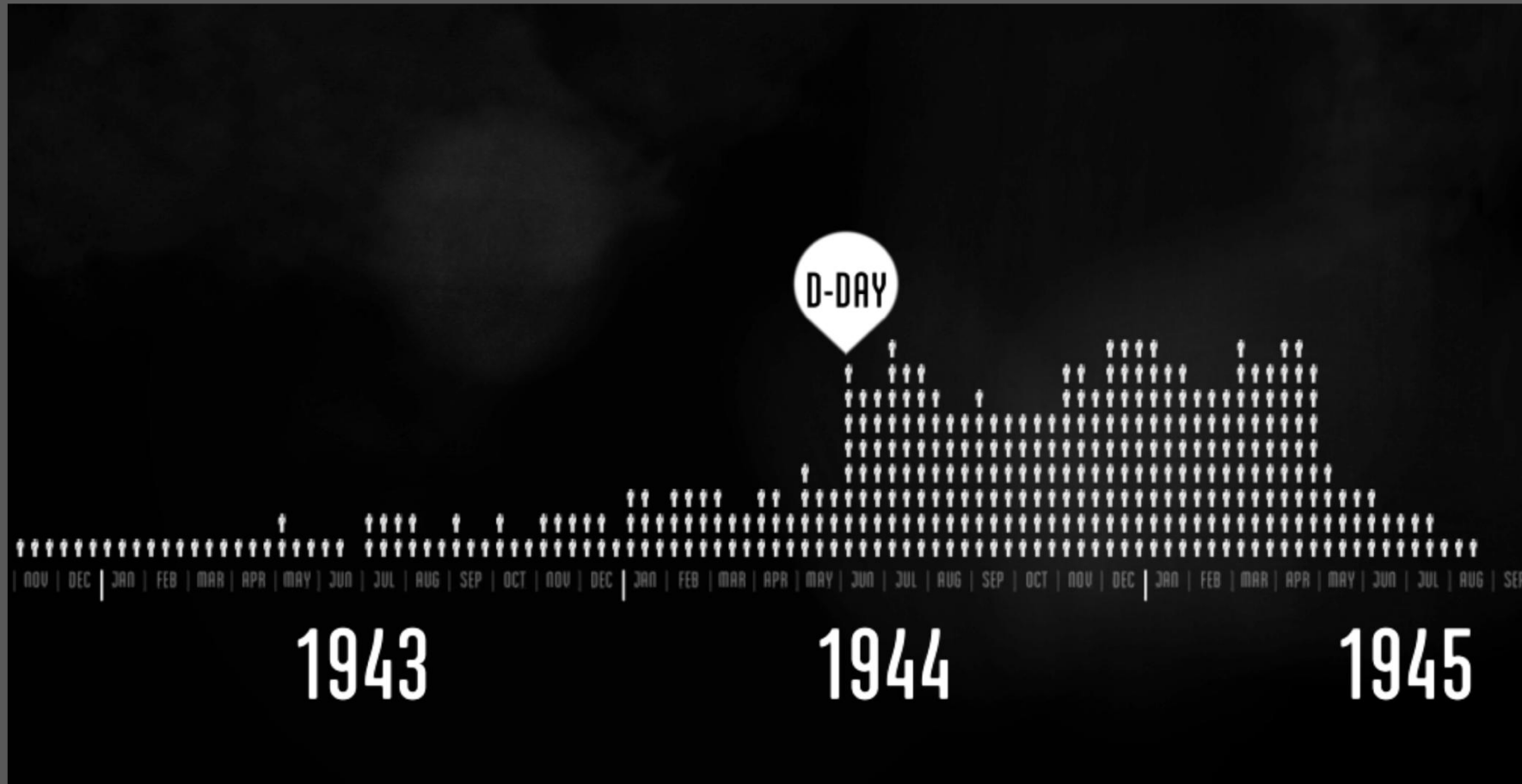
Sources: Standard and Poor's; Robert J. Shiller

Amanda Cox, Xaquín G.V. and David Leonhardt

Charting culture



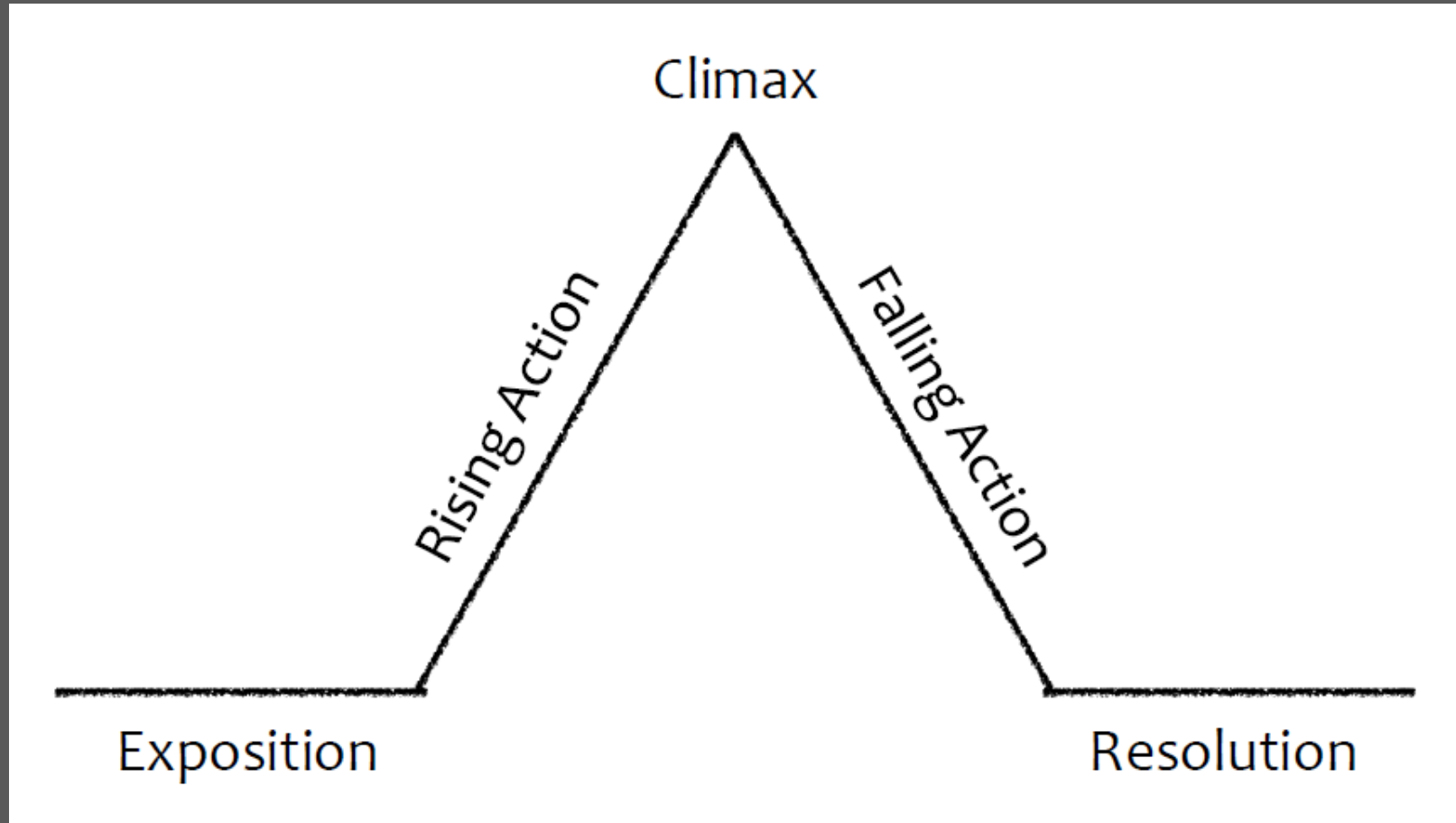
Fallen



How to give a talk

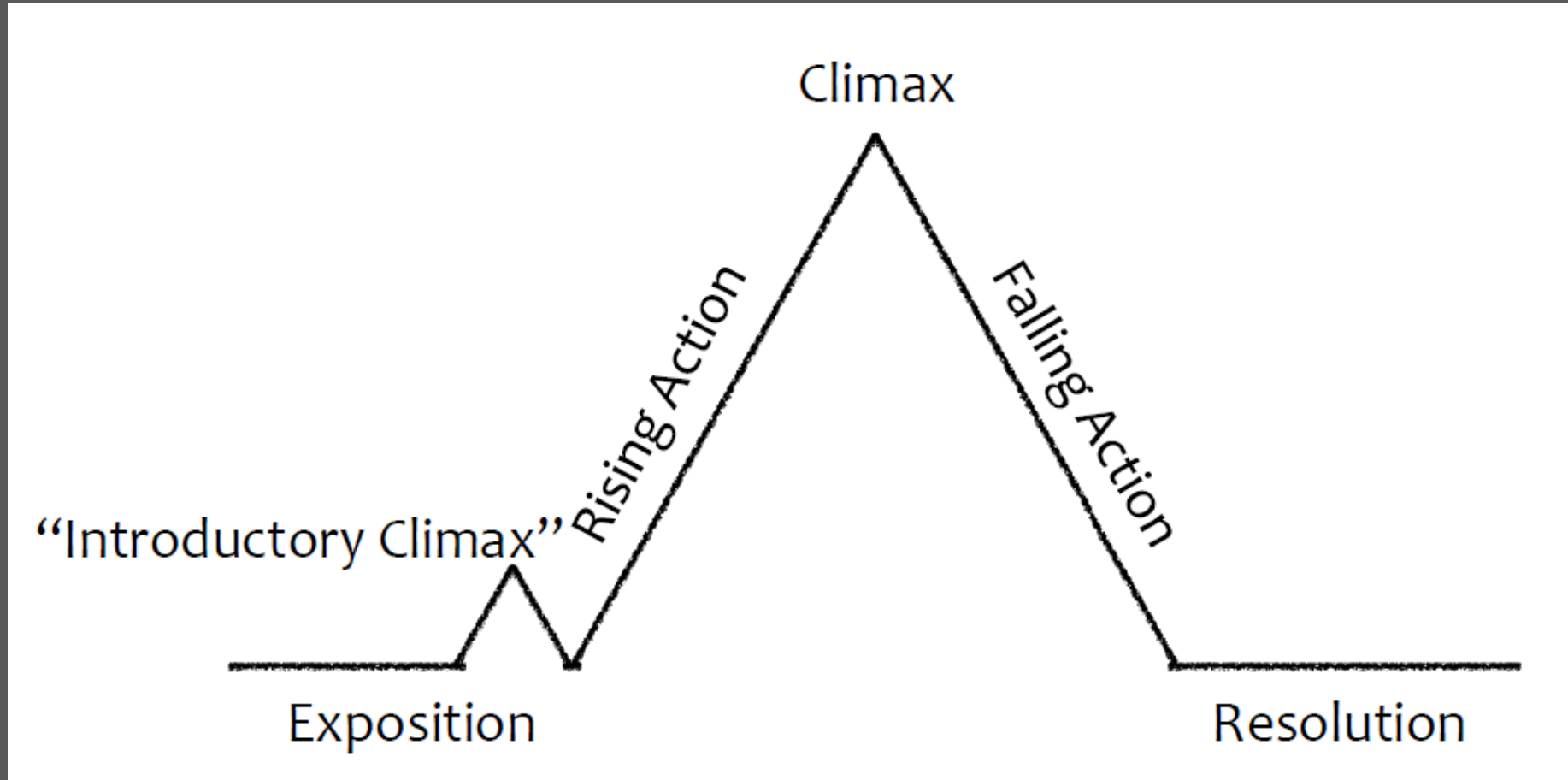
Structure, guidelines, tips, examples

Narrative structure



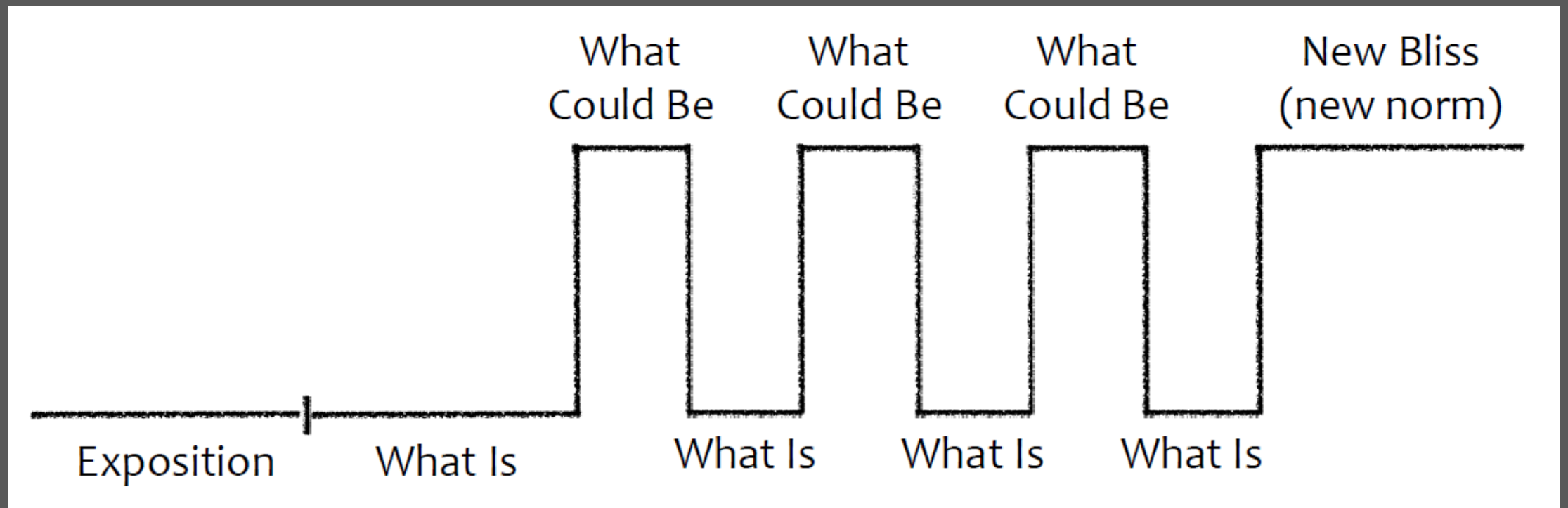
5 Stages Pyramid by Gustav Freytag

Narrative structure



6 Stages by Gustav Freytag

Narrative structure



Ted talk by Nancy Duarte: <https://www.youtube.com/watch?v=1nYFpuc2Umk>

TED's "Ten Commandments"

1. Dream big.

Strive to create the best talk you have ever given. Reveal something never seen before. Do something the audience will remember forever. Share an idea that could change the world.

TED's "Ten Commandments"

2. Show us the real you.

Share your passions, your dreams... and also your fears. Be vulnerable. Speak of failure as well as success.

TED's "Ten Commandments"

3. Make the complex plain.

Don't try to dazzle intellectually. Don't speak in abstractions. Explain! Give examples. Tell stories. Be specific.

TED's "Ten Commandments"

4. Connect with people's emotions.

Make us laugh! Make us cry!

TED's "Ten Commandments"

5. Don't flaunt your ego.

Don't boast. It's the surest way to switch everyone off.

TED's "Ten Commandments"

6. No selling from the stage!

Unless we have specifically asked you to, do not talk about your company or organization. And don't even think about pitching your products or services or asking for funding from stage.

TED's "Ten Commandments"

7. Feel free to comment on other speakers' talks, to praise or to criticize.

Controversy energizes! Enthusiastic endorsement is powerful!

TED's "Ten Commandments"

8. Don't read your talk. Notes are fine.

But if the choice is between reading or
rambling, then read!

TED's "Ten Commandments"

9. End your talk on time.

Doing otherwise is to steal time from the people that follow you. We won't allow it.

TED's "Ten Commandments"

10. Rehearse your talk

in front of a trusted friend ... for timing, for clarity, for impact.

Presentation tips

- Practice, practice, practice.
- Watch lots of talks for good role models.
- Breathe.
- Vary tone & rate of speaking.
- Take pauses. Avoid filler words (um, ah, uh).
- Never apologize.
- Always have a clear message.
- Make sure to have your “ah ha!” moment.
- Know your audience.

Example talks

- Terry Moore: How to tie your shoes
https://www.ted.com/talks/terry_moore_how_to_tie_your_shoes
- Hans Rosling's 200 Countries, 200 Years, 4 Minutes
<https://www.youtube.com/watch?v=jbkSRLYSojo>
- Semir Zeki: The neurobiology of beauty
<https://www.youtube.com/watch?v=NlzanAw0RP4>

Assignments

- **Formative:**

Evaluation of data visualisations

- **Summative:**

Explorable visual data stories (AE2)

“... The project should demonstrate mature visual and interaction design, showcasing iterative improvement and minimal usability issues by incorporating user feedback...”

Questions?